

Strings and languages

Strings

Strings play an important role in testing. A string serves as a test input. Examples: 1011; AaBc; “Hello world”.

A collection of strings also forms a language. For example, a set of all strings consisting of zeros and ones is the language of binary numbers. In this section we provide a brief introduction to strings and languages.

Alphabet

A collection of symbols is known as an **alphabet**. We use an upper case letter such as X and Y to denote alphabets.

Though alphabets can be infinite, we are concerned only with finite alphabets. For example, $X = \{0, 1\}$ is an alphabet consisting of two symbols 0 and 1. Another alphabet is $Y = \{\text{dog}, \text{cat}, \text{horse}, \text{lion}\}$ that consists of four symbols ``dog'', ``cat'', ``horse'', and ``lion''.

Strings over an Alphabet

A string over an alphabet X is any sequence of zero or more symbols that belong to X . For example, 0110 is a string over the alphabet $\{0, 1\}$. Also, **dog cat dog dog lion** is a string over the alphabet $\{\text{dog}, \text{cat}, \text{horse}, \text{lion}\}$.

We will use lower case letters such as p, q, r to denote strings. The length of a string is the number of symbols in that string. Given a string s , we denote its length by $|s|$. Thus $|1011|=4$ and $|\text{dog cat dog}|=3$. A string of length 0, also known as an **empty string**, is denoted by ε .

Note that ε denotes an empty string and also stands for “element of” when used with sets.

String concatenation

Let s_1 and s_2 be two strings over alphabet X . We write $s_1.s_2$ to denote the **concatenation** of strings s_1 and s_2 .

For example, given the alphabet $X=\{0, 1\}$, and two strings 011 and 101 over X , we obtain $011.101=011101$. It is easy to see that $|s_1.s_2|=|s_1|+|s_2|$. Also, for any string s , we have $s.\varepsilon=s$ and $\varepsilon.s=s$.

Languages

A set L of strings over an alphabet X is known as a **language**. A language can be finite or infinite.

The following sets are finite languages over the binary alphabet $\{0, 1\}$:

$\emptyset$: The empty set

$\{\epsilon\}$: A language consisting only of one string of length zero

$\{00, 11, 0101\}$: A language containing three strings

Regular expressions

Given a finite alphabet X , the following are **regular expressions** over X :

If a belongs to X , then a is a regular expression that denotes the set $\{a\}$.

Let r_1 and r_2 be two regular expressions over the alphabet X that denote, respectively, sets L_1 and L_2 . Then $r_1.r_2$ is a regular expression that denotes the set $L_1.L_2$.

Regular expressions (contd.)

If r is a regular expression that denotes the set L then r^+ is a regular expression that denotes the set obtained by concatenating L with itself one or more times also written as L^+ . Also, r^* known as the Kleene closure of r , is a regular expression. If r denotes the set L then r^* denotes the set $\{\epsilon\} \cup L^+$.

If r_1 and r_2 are regular expressions that denote, respectively, sets L_1 and L_2 , then $r_1 r_2$ is also a regular expression that denotes the set $L_1 \cup L_2$.